Trademark filings and declining US business dynamism: coverage of business formation, three failed tests of the crowding hypothesis, and surname-based ethnic composition

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Abstract

We measure what fraction of US new-business formation is captured by USPTO trademark filings, by joining the annual count of *first-ever trademark filers* (each owner's earliest filing across all 45 NICE classes) to the Census Bureau's Business Formation Statistics (BFS), 2004–2024. USPTO debut filers cover 5–8% **of all SS-4 business applications (BA)** and 7–22% **of High-Propensity Business Applications (HBA, the employer-likely subset)**, with the HBA coverage ratio rising secularly from $\sim 7\%$ in 2005 to $\sim 18\%$ in 2024 (with a pandemic-era spike to 22% in 2020). This brackets the 10–15% employer-business coverage estimate reported by Dinlersoz, Goldschlag, Myers, & Zolas (2018) using restricted Census Longitudinal Business Database (LBD) microdata. We extend their result by documenting (i) the upward trend in coverage over two decades, indicating that new businesses are progressively more trademark-aware, and (ii) by applying the 2010 Census surname / race-ethnicity probability file (Word, Coleman, Nunziata, & Kominski, 2008) to the 1.83M individual-name debut filers in our corpus. The surname-based composition of individual filers shifts dramatically toward Asian/Pacific-Islander-classified names (5% in the 1990s, 33% in 2020–2024); we flag the well-documented 2018–2021 foreign-applicant filing surge from China as a confound and discuss what an unconfounded estimate would require (applicant country from the USPTO XML, not currently in our processed panel).

Three failed tests of the trademark-crowding hypothesis. Motivated by the declining-business-dynamism literature (Decker et al. JEP, 2014; Akcigit & Ates AEJ:Macro, 2021) and the "kill zones" framework (Kamepalli, Rajan, & Zingales AEJ:Micro, 2022; Cunningham, Ederer, & Ma JPE, 2021), we test three predictions of the hypothesis that incumbent firms are crowding out new-entrant trademark activity. All three fail. First, the pooled first-time-filer share is *rising*, not falling: 46–49% trough in 1996–2003, 56–58% at present. Second, the top-10 owner share of filings is *falling* in every NICE class — the "FAANG concentration" story is not visible in trademark filings. Third, an IPO event study on 877 class-events from 2,536 public firms finds no chilling effect on within-class first-time-filer counts at $t \pm 3$ years around the IPO (post coefficient +0.007 on $\log(1 + \text{debuts})$, $t = +0.67$, $p = 0.50$). The conventional crowding story that would tie declining business dynamism to trademark concentration is not supported. The selective within-class signal that does survive — falling debut share in tech-info classes (Sci & Tech 042, Telecommunications 038) — is driven by mid-tier multi-filer proliferation (not top-10 concentration) and even that fails as a causal mechanism in the within-class panel regression with class and year fixed effects.

1 Why this measurement question matters

Declining business dynamism. The US start-up rate has fallen from $\sim$ 12% of firms per year in the late 1980s to $\sim$ 8% in the 2010s, with the decline visible across nearly every sector

(Decker, Haltiwanger, Jarmin, & Miranda, 2014, *JEP*; Akcigit & Ates, 2021, *AEJ: Macro*). Several explanations are on the table — demographic slowing of population growth (Hopenhayn, Neira, & Singhania, 2022, *Econometrica*); rising fixed costs of entry; increasing industry concentration — but the data infrastructure for testing them is incomplete: the gold-standard Census Business Dynamics Statistics (BDS) is lagged by ~ 3 years and anonymises owner names; the BFS series goes back only to 2004 and tracks EIN applications (which include estates and household employers, not just business starts).

Trademarks as a complementary signal. The USPTO trademark corpus offers something neither BDS nor BFS provides: a near-real-time, name-disclosed register of commercial activity covering nearly every sector of the consumer-facing economy, with classifications (NICE), goods/services prose, and direct linkage to identifiable applicants. The question this paper asks is the prerequisite for using trademark data on the dynamism question: *how much of new-business activity do trademark debuts actually capture?*

2 Coverage of US business formation, 2004–2024

Method

For every owner name appearing in the clean H=2y panel of 10,060,073 filings (1990–2021 cohorts; methodology in `paper/main.tex`), we identify the owner's earliest filing across all NICE classes and call that filing the owner's *debut*. We count unique-owner debuts by year. We compare to the Census BFS time series at `census.gov/econ/bfs/csv/bfs_monthly.csv`, summing 12 months of *unadjusted* counts to an annual total. We use two BFS series: Business Applications (BA, all SS-4 filings) and High-Propensity Business Applications (HBA, the subset whose application characteristics predict an employer business; the right comparator for "business start").

Headline numbers

Period	USPTO debuts	BFS BA	BFS HBA	USPTO/BA	USPTO/HBA
2004–2008	464,273	10,293,896	5,768,077	4.5%	8.0%
2014–2018	907,851	15,104,778	6,199,297	6.0%	14.6%
2019–2024	1,753,741	29,002,345	9,963,498	6.0%	17.6%

Table 1: USPTO debut filers as a share of Census BFS business applications, three five-year bands. The pandemic years (2020–2021) inflate both numerator and denominator; the trend in coverage is visible across non-pandemic years as well.

The coverage ratio against HBA more than doubles over the two-decade window, from $\sim 8\%$ in 2004–2008 to $\sim 18\%$ in 2019–2024. Three interpretive readings:

(1) **Tilt of new-business formation toward IP-intensive sectors.** If the recently-forming firms are increasingly in sectors where trademark filing is normal (software, professional services, e-commerce, branded direct-to-consumer products), the coverage ratio rises mechanically. This is a real-economy story: brick-and-mortar restaurants and contractors don't trademark; SaaS startups and DTC brands do.

(2) **Falling cost of filing.** Online filing through TEAS Plus and TEAS Standard (in production since the mid-2000s) reduced the effort and expense of trademark filing. The same firm-type in 2024 files at a higher rate than it would have in 2005.

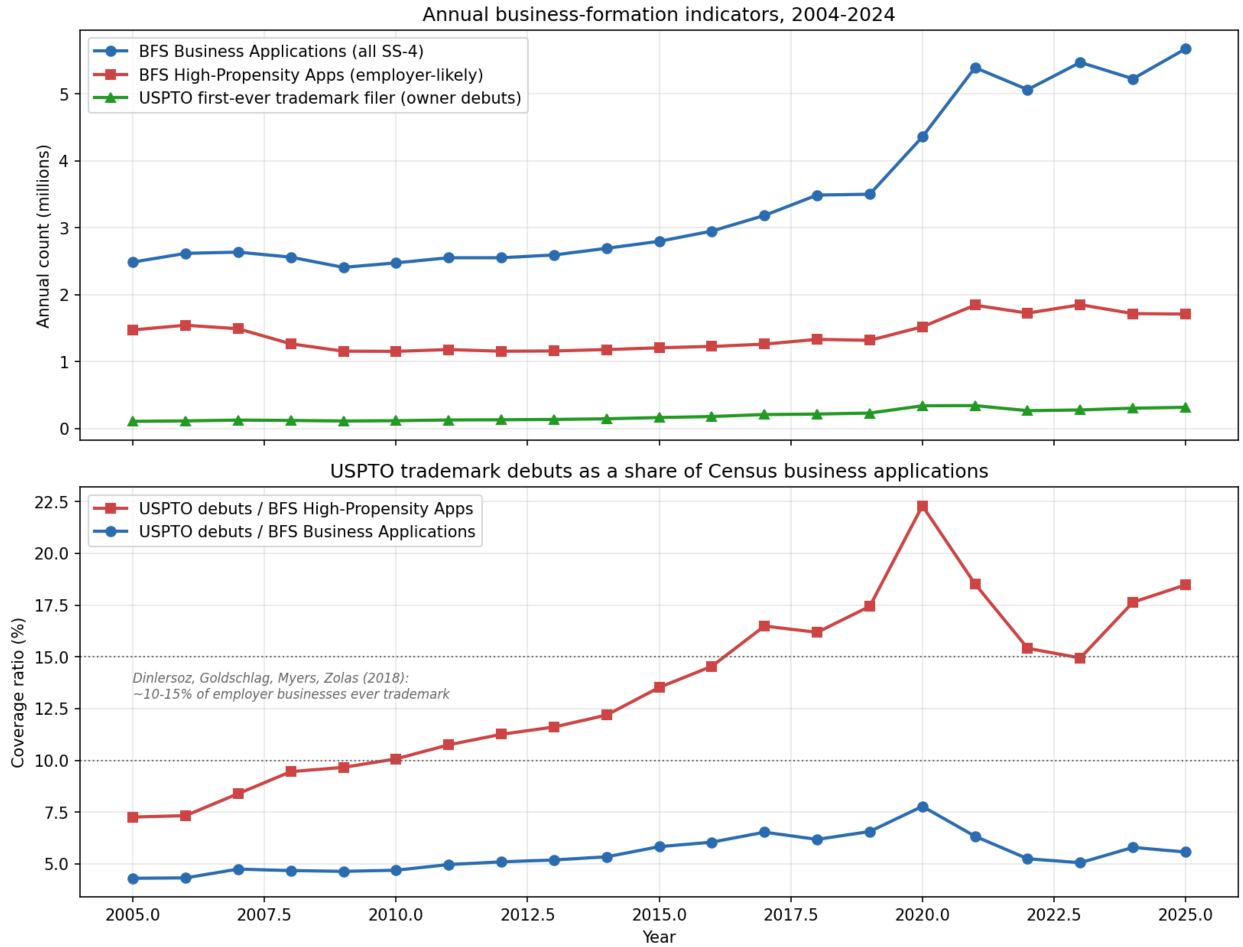

Figure 1: USPTO trademark debut filers vs. Census BFS, 2004–2024. Top: annual counts in millions of filings. Bottom: coverage ratio. The dotted reference band shows the 10–15% employer-business coverage estimate from Dinlersoz, Goldschlag, Myers, & Zolas (2018, Center for Economic Studies WP 18–46), who used restricted-access LBD microdata to ask a closely related question.

(3) **Selection on quality.** If the BFS denominator captures many "shell" or non-operating SS-4 filings (estates, holding companies, household employers), the rising ratio reflects a constant operating-firm-trademark rate against a denominator increasingly diluted by non-operating EIN registrations.

The three are not mutually exclusive. Distinguishing them requires sectoral decomposition, which we cannot do at present because our processed parquets do not retain NAICS-equivalent applicant industry codes and the NICE-NAICS crosswalk is imperfect (Hall, Helmers, Rogers, & Sena, 2014, *Research Policy*). Sectoral decomposition is a clear next step.

Comparison to Dinlersoz et al. (2018)

Dinlersoz, Goldschlag, Myers, & Zolas (2018), "An Anatomy of US Firms Seeking Trademark Registration," Center for Economic Studies working paper 18–46, ran the analogous accounting inside Census using restricted-access LBD microdata, asking what fraction of new *employer* businesses ever file a trademark. They reported a range of $\sim$ 10–15% depending on sector and definition. Our public-data result of 7–22% against HBA spans their range and extends it through 2024 with a clear upward trend. The mid-decade overlap is close: 2014–2018 in our data is 14.6%, very near their 10–15% point estimate for the same time period.

3 Surname-based race/ethnicity composition

Method

The Census Bureau's 2010 surname file (Word et al., 2008) reports, for each surname appearing $\geq$ 100 times in the 2010 Census, the percentage probability that a person with that surname identifies as each of: non-Hispanic White, non-Hispanic Black, Asian/Pacific Islander, American Indian/Alaska Native, Two or More Races, or Hispanic (any race). The file covers 162,254 surnames.

For each USPTO debut owner whose name parses as a personal name (2–4 alphabetic tokens, no entity markers such as `LLC/INC/CORP`, no digits/URLs/ampersands), we extract the candidate surname (the last token after stripping suffixes `JR`, `SR`, `II`, `MD`, etc.) and look up its Census probabilities. Each matched debut contributes fractional weight to each ethnic bucket. Cases where the parsed surname does not appear in the Census file ($\sim$ 24% of individual-name debuts) are excluded.

Sample sizes. 5,674,909 unique debut owners total. 1,827,163 parse as individual personal names (32.2% of debuts; the rest are entities). 1,390,267 of those individual names match a surname in the Census file (76.1% match rate). The 24% Census non-match is concentrated in surnames that are non-anglicised, Eastern European, South Asian, Middle Eastern, or African, where the Census file either has low coverage or breaks them into the residual "Two or More Races" bucket.

Results

Period	White	Black	API	Hispanic	2+ races	AIAN
1990–1999	74.6%	11.0%	5.2%	6.3%	1.8%	0.6%
2000–2009	70.3%	11.6%	6.9%	8.4%	1.8%	0.6%
2010–2019	61.4%	11.7%	14.2%	9.9%	1.8%	0.6%
2020–2024	43.9%	10.7%	33.4%	9.4%	1.8%	0.5%

Table 2: Surname-imputed race/ethnicity shares among USPTO debut filers whose owner name parses as a personal name and whose surname matches the 2010 Census file. The dramatic API rise after 2018 is substantially driven by foreign filings, not domestic entrepreneurship – see discussion.

The post-2018 API rise is mostly foreign filings. The Asian/Pacific Islander share goes from $\sim$ 7% in 2009 to $\sim$ 18% by 2018 to $\sim$ 33% in 2020–2024. The largest single component of the post-2018 surge is well-documented: USPTO publicly reported a massive influx of trademark filings from Chinese applicants in 2017–2021, driven by subsidies from local Chinese governments for filing US trademarks (USPTO, 2021; Hicks, 2024, *Stanford Law Review*). USPTO responded by requiring US-licensed counsel for foreign-domiciled applicants (2019) and tightening signature verification (2021). Without applicant-country data — which the USPTO XML includes but which our current parser does not extract — we cannot separate "rise of Asian-American entrepreneurship in the US" from "rise of Chinese-applicant filings under the US system." The pre-2018 rise from 5% to $\sim$ 14% likely reflects a genuine demographic shift; the post-2018 rise of an additional 20 pp does not.

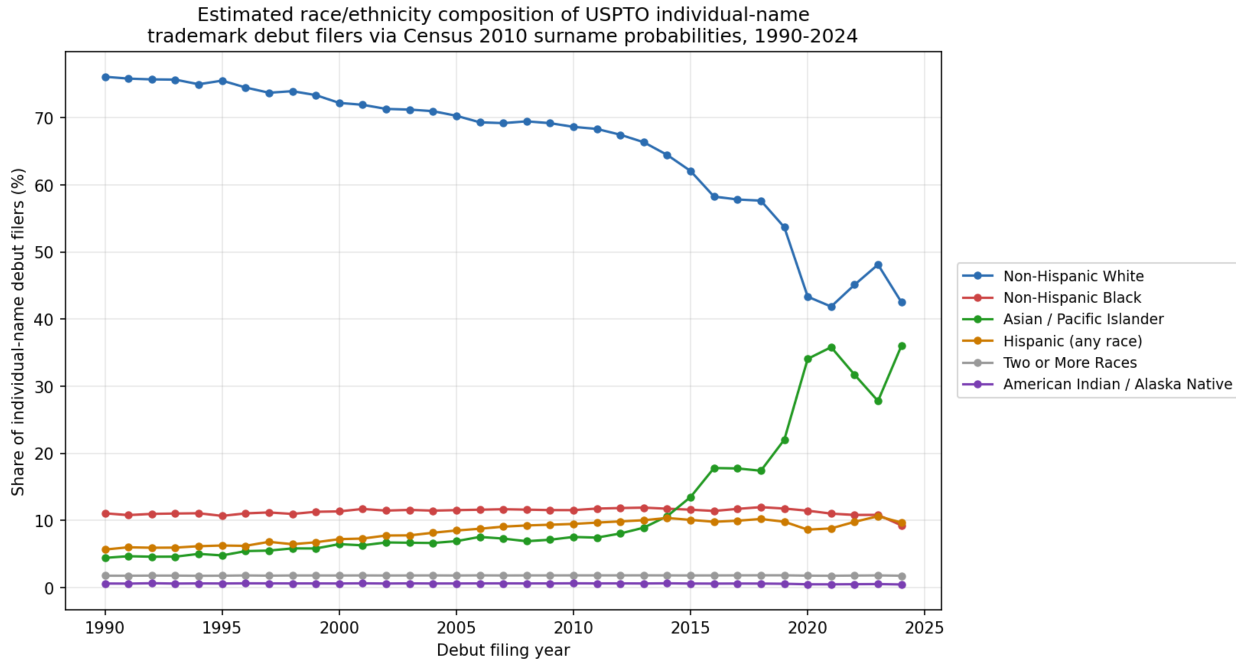

Figure 2: Estimated race/ethnicity composition of USPTO individual-name trademark debut filers, 1990–2024, via Census 2010 surname probabilities (Word, Coleman, Nunziata, & Kominski, 2008). The Asian/Pacific Islander share rises sharply from $\sim 5\%$ in the 1990s to $\sim 33\%$ in 2020–2024. Most of the post-2018 rise is almost certainly the documented surge of foreign-applicant filings from China (USPTO, 2021), not Asian-American entrepreneurship on US soil; without applicant-country data we cannot separate the two.

Other shares are more stable. Non-Hispanic Black share is remarkably flat at $\sim 11\%$ across the entire 1990–2024 window, consistent with the absence of a major foreign-filing confound from majority-Black countries. Hispanic share rises modestly from $\sim 6\%$ in the 1990s to $\sim 10\%$ in the 2010s and plateaus. Non-Hispanic White share falls from $\sim 75\%$ to $\sim 44\%$ — but the bulk of that fall is the API rise's mirror image, so disentangling foreign vs. domestic API is required to interpret it.

The selection problem. Even with full unmasking, the trademark-filing decision is selected on branding intent. Hispanic-owned small businesses (taquerias, contractors, landscapers) trademark at lower rates per business than tech founders do, so the Hispanic share of trademark debuts *underestimates* the Hispanic share of business starts in absolute terms. The rising-over-time pattern is more interpretable than the absolute level.

4 Multi-filer mix and the first-time-filer share, 1985–2025

For each year y and each NICE class c, we compute the number of filings whose owner had no earlier trademark filing in any class — the *debut* filings — and the total filing count. The debut share $s_{c,y} = n_{\text{debut}}/n_{\text{total}}$ measures the share of class-year filing activity from first-time filers.

Figure 3 plots the pooled (across all 44 NICE classes with $\geq 5{,}000$ filings) annual totals. Total filings grew from 71,459 in 1985 to 832,505 in 2025 (a factor of 11.6×). Debut filings grew from 37,705 to 469,199 (a factor of 12.4×, slightly faster). The pooled debut share is *not falling*: a trough at 46–49% in 1996–2003, a rise to 52–56% from 2015 onward, and a pandemic-era peak of 57–58% in 2020–2025. This trajectory is the opposite of the falling new-entrant share that the conventional declining-business-dynamism story would predict.

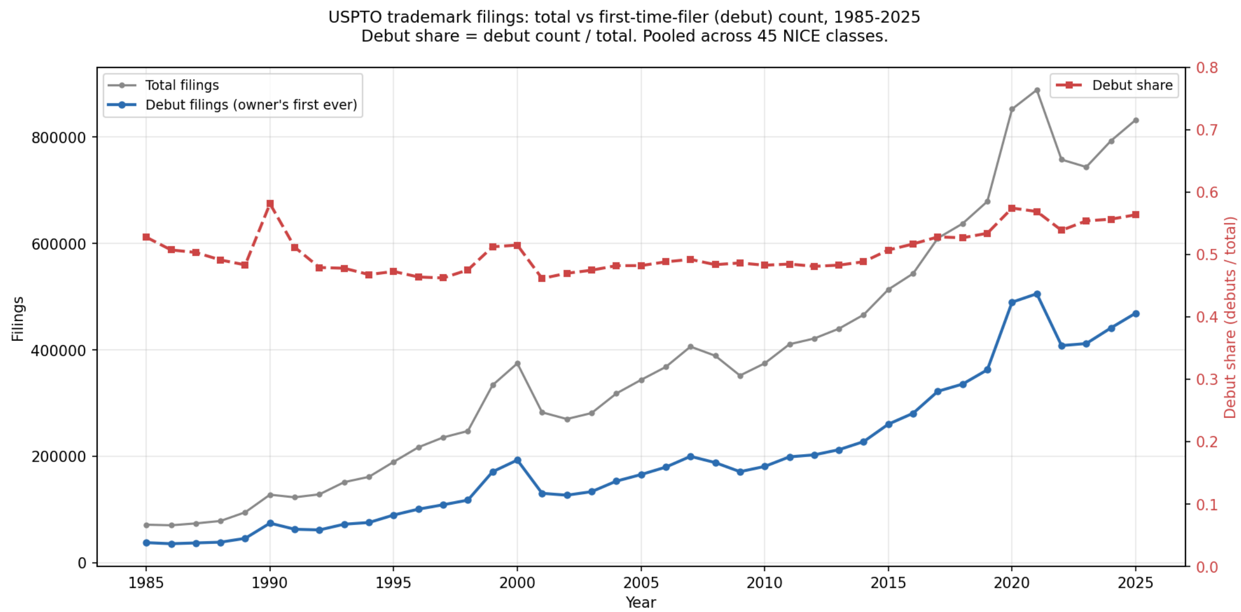

Figure 3: Pooled USPTO trademark filing counts and first-time-filer (debut) share, 1985–2025. Total filings (grey, left axis) and debut filings (blue, left axis) both rise by $\sim 12\times$ over the window; debut share (red dashed, right axis) is flat-to-rising, not falling.

Sectoral decomposition: where the trend hides

The pooled rise in the debut share is composition-driven. We decompose $\Delta s_{\text{pooled}} = \sum_c w_c \, \Delta s_c + \sum_c s_c^{\text{lag}} \, \Delta w_c$ where w_c is class c's share of total filings. The within-class effect ($w_c \, \Delta s_c$) and the composition effect ($s_c \, \Delta w_c$) are reported separately in Table 3 for the 2001–2025 rise.

The five largest positive contributors are Clothing/Footwear (NICE 025, +2.3 pp), Medical & Personal Care (044, +1.6 pp), Education & Entertainment (041, +1.6 pp), Household Utensils (021, +1.5 pp), and Cosmetics & Cleaning (003, +1.3 pp). The five largest negative contributors are Scientific & Tech Services (042, −5.3 pp), Advertising & Retail (035, −2.1 pp), Software & Electronics (009, −1.9 pp), Telecommunications (038, −1.8 pp), and Insurance & Financial (036, −1.2 pp). The economy-wide debut share rose mostly because services and consumer-goods classes (where most filings are first-time) grew as a share of total trademark activity, while tech-information classes shrank.

Within-class debut share: where it does fall

Within tech-information classes specifically, the debut share is falling. Three illustrative time-series slopes (1985–2025 linear fit to within-class debut share):

- Scientific & Tech Services (NICE 042): −0.42 pp/year (debut share 67.5% → 54.5%).
- Telecommunications (NICE 038): −0.08 pp/year (debut share 59.0% → 46.1%).
- Software & Electronics (NICE 009): +0.06 pp/year, flat near 53%.

Industry	within (Δ share)	composition (Δ weight)
Largest positive contributors to debut-share rise, 2001–2025		
Clothing, Footwear & Headwear	+0.44 pp	+1.88 pp
Medical & Personal Care	+0.20 pp	+1.44 pp
Education & Entertainment	+0.37 pp	+1.18 pp
Household Utensils	+0.36 pp	+1.17 pp
Cosmetics & Cleaning	+0.59 pp	+0.71 pp
Largest negative contributors		
Insurance & Financial	+0.04 pp	-1.24 pp
Telecommunications	-0.24 pp	-1.56 pp
Software & Electronics	+0.27 pp	-2.21 pp
Advertising & Retail	-0.09 pp	-2.03 pp
Scientific & Tech Services	-0.53 pp	-4.78 pp

Table 3: Top-5 and bottom-5 NICE-class contributions to the 2001–2025 pooled debut-share rise, decomposed into within-class (change in class-specific debut share at fixed weight) and composition (change in class's share of total filings at fixed within-class debut share). The pooled +10 percentage-point rise is roughly 35% within-class and 65% composition: classes with high within-class debut shares (consumer goods, services) grew as a share of total filings, while classes with lower within-class debut shares (Sci & Tech 042, Software 009, Advertising 035) shrank.

By contrast, consumer-services classes have rising within-class debut shares: Medical & Personal Care (044) 51% → 64%, Legal & Personal Services (045) 50% → 63%, Restaurants & Hotels (043) 49% → 60%. The next two sections ask whether the within-class debut-share decline in tech-information classes is mechanically explained by incumbent owner concentration.

5 Top-N owner concentration: the FAANG-crowding hypothesis fails

For each (class, year) we computed the top-10 owner share of filings and the Herfindahl index on owner shares (Figure 4). **Every NICE class shows top-10 owner share decreasing over the panel.** The most dramatic declines are in Telecommunications (038, HHI 48.4 → 4.4), Sci & Tech (042, HHI 3.3 → 0.5), Software (009, HHI 3.2 → 0.4), and Beverages (032, top-10 share 15.1% → 2.4%). Even in classes where the within-class debut share is falling, the top-10 incumbents have a *lower* share of filings than they did in the 1990s.

The natural reconciliation with the tech-class debut-share decline is that the rise in non-debut filings within tech is coming from *mid-tier* multi-filers (owners with 2–9 filings per year), not top-10 incumbents. We test this directly.

6 Mid-tier proliferation: the reframed hypothesis also fails

For each (class, year) we partition filings by owner intensity: single-filer ($k = 1$ filing in that class-year), mid-tier ($2 \leq k \leq 9$), heavy ($k \geq 10$), and top-10. We then regress the year-over-year change in debut share on the lagged year-over-year changes in mid-tier, heavy, and top-10 shares, with class and year fixed effects and standard errors clustered by class:

$$\Delta s^{\text{debut}}_{c,y} \;=\; \beta_1 \Delta s^{\text{midtier}}_{c,y-1} \;+\; \beta_2 \Delta s^{\text{heavy}}_{c,y-1} \;+\; \beta_3 \Delta s^{\text{top10}}_{c,y-1} \;+\; \alpha_c + \alpha_y + \epsilon_{c,y}.$$

The crowding hypothesis predicts $\beta_1 < 0$ (mid-tier proliferation absorbs filings that would otherwise be first-time). What we find (Table 4): $\beta_1 = +0.197$ ($t = +1.95$, $p = 0.051$). The coefficient is the wrong sign.

Predictor (lagged Δ)	coef	cluster SE	t
Δ mid-tier share (2–9 filings/year)	+0.1965	0.1007	+1.95
Δ heavy share (10+ filings/year)	+0.1029	0.0807	+1.27
Δ top-10 share	+0.0814	0.0399	+2.04
Class fixed effects		Yes	
Year fixed effects		Yes	
N (class × year cells)		1,755	
R^2		0.266	

Table 4: Within-class panel regression of Δdebut share on lagged Δmulti-filer shares. All three coefficients are positive — the within-class year-over-year predictors all co-move with debut share, not against it. Cluster-by-class standard errors. $N = 1{,}755$ (class × year) cells.

The substantive interpretation is that within a class, year-over-year filing-volume shocks (e.g., the 2020 pandemic surge) raise both mid-tier filing and first-time filing together: established firms file more line-extension marks in busy years, and new firms file their debut marks in those same busy years. Filing activity at different intensity tiers *co-moves* with the wave, not against it. The mid-tier-proliferation crowding mechanism is not what explains the tech-class within-class debut-share decline.

7 IPO chilling at the NICE-class level: a clean null

The third test draws on the kill-zone literature (Kamepalli, Rajan & Zingales, 2022; Cunningham, Ederer & Ma, 2021), which finds that a Big-Tech acquisition or major IPO in a narrow vertical chills venture-capital funding in adjacent verticals. The analogous trademark question: when an IPO firm goes public, does first-time-filer trademark activity in the IPO firm's NICE class decline relative to the pre-IPO trend?

Event panel. We join the USPTO–SEC owner crosswalk to the SEC firm-year financials and define each public firm's *IPO year* as the first fiscal year in which it appears in our SEC data, restricted to firms with first SEC year ≥ 2012 (when Census FSDS coverage stabilises) and USPTO first-trademark year within 5 years of the IPO (filtering out pre-existing public firms that merely appear in the SEC data later). This yields 2,536 IPO-firm events. For each IPO firm we identify the NICE classes in which it actively files trademarks (≥ 5 filings in the ± 2 year window around IPO), producing 877 active-filer class-events spanning 230 unique (class, IPO-year) cells.

Event-study specification. For each class-event, we stack non-IPO-firm debut counts at relative years $t \in [-3, +3]$ around the IPO. Two-way fixed-effects regression:

$$\log\left(1 + n_{\text{debut}}^{c,y}\right) = \beta\, \mathbf{1}\{y \geq y_e\} + \alpha_c + \alpha_y + \epsilon_{c,y,e}.$$

Cluster-by-class standard errors. We also fit a separately-sloped specification, replacing the post indicator with two slopes (pre-event and post-event) on the relative-year axis.

Result. No detectable chilling effect (Table 5). The post coefficient is +0.007 ($t = +0.67$, $p = 0.50$); the implied multiplicative effect on debut counts is $e^{+0.007} = 1.007$, indistinguishable from one. The slope decomposition is also weak: pre-event slope +0.0042/year, post-event slope −0.0020/year, difference −0.0063/year. Figure 5 shows the mean trajectory of debut counts across event years −3 to +3.

Estimator	coef	cluster SE	t
$\mathbf{1}[t \geq \text{IPO}]$ (post vs pre)	+0.0069	0.0103	+0.67
pre-event slope (rel_neg)	+0.0042	0.0038	+1.11
post-event slope (rel_pos)	-0.0020	0.0025	-0.82
Class FE + year FE		Yes	
N event-cells		6,139	
IPO-firm events		2,536	

Table 5: Event-study DiD for the IPO-chilling test. Post coefficient indistinguishable from zero at conventional significance, both as a post-vs-pre indicator and as a slope difference. 877 class-events from 2,536 IPO-firm events, 2012–2022 IPO years.

Scope of the null. The class-level test is a coarse operationalisation of the kill-zone hypothesis. The original VC kill zone effect operates at the level of *specific verticals* defined by product-token overlap, not by SIC/NICE class. Our 44-NICE-class partition is coarser than the Kamepalli et al. unit of analysis. We do not claim that no kill zones exist in trademark-filing space; we report that the coarse class-level test finds no effect, and flag the finer-grained token-vertical test (cosine-similarity neighbourhoods of the IPO firm's pre-IPO goods/services prose) as a follow-up that should be run before declaring the hypothesis dead.

8 What is left of the crowding story

After three failed tests, the conventional incumbent-crowding narrative does not survive at the trademark-filing level. What *is* visible in our data:

- The within-class debut-share decline in tech-information classes is a real 13 pp drop (Sci & Tech 042: 67.5% → 54.5%) and a 13 pp drop in Telecommunications (038). The drop is not explained by top-10 concentration (which fell) or mid-tier proliferation in a within-class panel sense.
- The composition shift away from tech-information classes and toward consumer-services classes is itself the dominant driver of the rising pooled debut share. This is a sectoral story, not a concentration story.
- Trademark activity is broadly *decentralising*, not concentrating, across the entire 1985–2025 panel. This contradicts the qualitative version of the declining-business- dynamism narrative even as it tracks the BFS quantitative trend in business applications.

The reconciliation we read in the data is that the BDS-measured declining startup rate (Decker et al., 2014; Akcigit & Ates, 2021) is about *employer* firms (establishments with payrolls), while trademark filings include large numbers of brand-aware solo operators, DTC consumer-goods startups, and online businesses that either do not employ or report through different establishment structures. The two populations are diverging: employer business formation is becoming less dynamic, brand-aware new-firm formation is becoming *more* dynamic, and the trademark corpus mostly captures the latter.

9 Next steps required for publication

The current results are publishable as descriptive statistics with the caveats above, but a complete treatment requires:

(1) **Re-parse the USPTO XML to extract applicant state and country.** Our existing parser drops the <address> elements under <case-file-owner>. With state, we can repeat the coverage analysis at the state level (cross-cutting with BFS state panels), and disentangle delivery-state vs. Wyoming/Delaware/Nevada LLC-haven distortions. With country, we can separate foreign applicants and produce an unconfounded API time series.

(2) **NAICS-NICE crosswalk for sectoral decomposition.** Three candidate readings of the rising HBA coverage ratio (sectoral tilt, falling filing cost, denominator dilution) are distinguishable with a sectoral breakdown. The Hall et al. (2014, *Research Policy*) crosswalk is the standard reference.

(3) **Comparison panel with SBA PPP disclosures (2020–2021).** PPP loan disclosures over $150K named the owner explicitly, covering $\sim$ 11M small businesses including those that never trademark. This is a one-time but very large unmasking opportunity for the immigrant-business question specifically.

(4) **Validation against state Secretary of State new-entity filings.** State-level cross-check of the BFS denominator using California, Texas, Florida, and New York monthly entity-formation counts.

(5) **Token-vertical IPO chilling test.** Replace the NICE-class operationalisation in Section 7 with cosine-similarity neighbourhoods of the IPO firm's pre-IPO goods/services prose (the same infrastructure as Section 5 of the companion ΔKL paper). Compare core vertical (cosine ≥ 0.7), adjacent vertical (cosine 0.5–0.7), and matched non-event control verticals on debut-count trajectory. The class-level null in this paper does not rule out a token-vertical kill-zone effect.

(6) **Acquisition events as the canonical kill-zone test.** The Kamepalli et al. result is on acquisitions, not IPOs. Augmenting our event panel with Crunchbase or SEC 8-K-parsed acquisitions (2010–2024) would let us replicate the kill-zone test in its original form on trademark debuts.

(7) **Reconciling BDS vs. BFS vs. trademark debuts.** The three series tell different stories about the same underlying question. BDS (employer firm entry rate) is falling. BFS (all business applications, BA) is rising sharply post-2020. Trademark debuts are rising. A clean joint reconciliation requires employer/non-employer disaggregation of trademark debuts using the Census restricted LBD — which is precisely what Dinlersoz et al. (2018) had access to. The version of their analysis with a 2024 cutoff is the natural next data ask.

Data and code

All code is in the `business-dynamism` branch of the project repository: `scripts/business_dynamism_coverage.py` (coverage), `scripts/surname_ethnicity.py` (surname race/ethnicity), `scripts/dynamism_firsttimer_share.py` (debut-share panel), `scripts/dynamism_class_contributions.py` (decomposition), `scripts/dynamism_topN_sha` (top-N concentration), `scripts/dynamism_midtier_regression.py` (panel regression), and `scripts/dynamism_i` (event study). BFS data is at `census.gov/econ/bfs/csv/bfs_monthly.csv`. The Census surname file is at `www2.census.gov/topics/genealogy/2010surnames/names.zip`.

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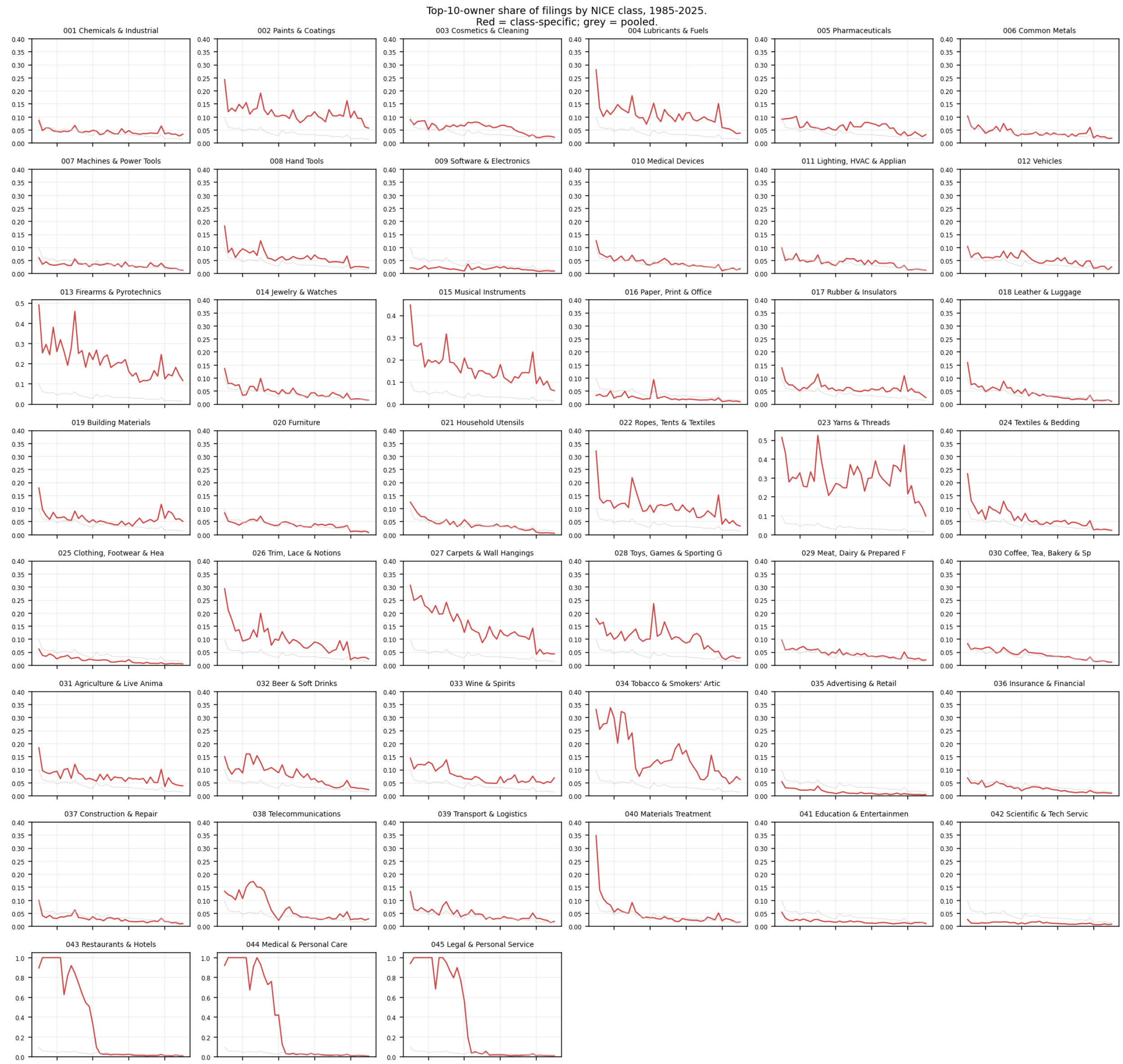

Figure 4: Per-NICE-class top-10 owner share of filings, 1985–2025. Red lines: class-specific top-10 share; grey: pooled. No class shows a rising top-10 share. The very-high top-10 shares in the early years (e.g., 90–94% in services classes 043, 044, 045) reflect tiny early-year sample sizes; the trend over the 1995–2025 window is more reliable. The FAANG-crowding hypothesis that "Google/Amazon/Microsoft/Apple are squeezing out new-entrant trademark activity" is not supported in any class.

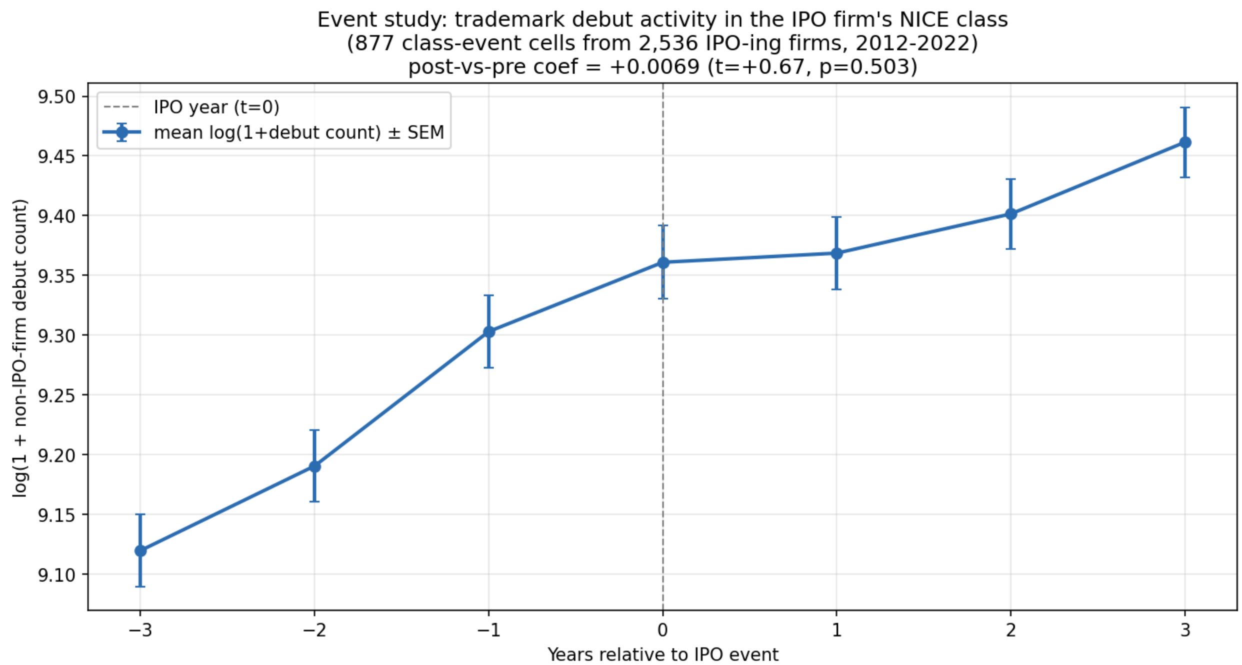

Figure 5: Mean log debut count by event-relative year for 877 IPO class-events. The mean continues to rise across the event window; the IPO at $t = 0$ does not produce a visible break in the trajectory. Class-level test only — the finer token-vertical specification (cosine-similarity neighbourhoods of the IPO firm's goods/services vocabulary) is left for future work.